

Coarsening Confers Robustness: Causal Bayesian Optimization under DAG Misspecification

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Abstract

Causal Bayesian optimization (CBO) uses a causal graph to select intervention sets and construct priors over the interventional response. A wrong graph can therefore bias a prior or remove an optimal arm. We replace the full graph by a partition and its cluster-level quotient. Any optimizer whose graph-dependent components factor through this input has the same trajectory for all fine graphs with the same quotient. We instantiate the contract for three families of the field: arm-based, dynamic, and model-based CBO. Under full intervention support and cluster-contained confounding, coarsening preserves the best attainable value. Intervention constraints or confounding outside a cluster can create a positive action-space gap. We characterize this gap, show its linear contribution to incumbent regret, and recover fine actions by adaptive refinement. Experiments isolate prior and arm-set corruption, verify exact quotient invariance, and test all three families on their standard benchmarks.

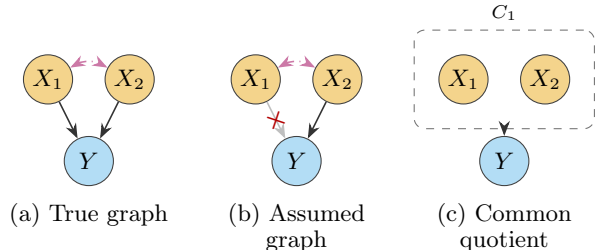


Figure 1: PARALLELPARENT. Deleting $X_1 \rightarrow Y$ changes the fine search space but not the quotient because $C_1 \rightarrow Y$ remains through $X_2 \rightarrow Y$. Orange nodes are manipulable, blue is the target, and the dash-dotted edge denotes latent confounding.

from the truth. Our premise is simple: commit only to structural distinctions that are reliable.

Wrong edges are costly: two-variable example. Let X_1 and X_2 be confounded parents of Y , with

$$Y = \lambda(X_1 - 2)^2 + (X_2 + 2)^2 + \varepsilon, \quad y^* = 0.$$

Deleting $X_1 \rightarrow Y$ from the assumed graph removes every X_1 -containing fine arm. The best remaining intervention satisfies

$$\min_{x_2} \mathbb{E}[Y \mid \text{do}(X_2 = x_2)] = 4.25\lambda.$$

For $C_1 = \{X_1, X_2\}$, however, both assumptions induce the same quotient $C_1 \rightarrow Y$ (Fig. 1). A quotient method therefore follows the same trajectory under both assumptions.

Quotient CBO. Full-DAG methods and online structure learning have both been studied [Mukherjee et al., 2024, Branchini et al., 2023]. We study the middle ground. The practitioner supplies a partition Π and a cluster DAG (C-DAG) over its clusters. One cluster-level judgement replaces several edge-level judgements. QCBO applies CBO’s minimal-intervention-set rule to this quotient and uses identifiable cluster effects as prior means. Cluster-level

1 Introduction

Causal Bayesian optimization [Aglietti et al., 2020] searches over interventions and uses observational data with a causal graph to build priors over the interventional response. CBO and its descendants [Aglietti et al., 2021, 2023, Sussex et al., 2023] usually assume that the full graph is correct. Yet changing only the assumed graph can substantially change their performance [Anonymous, 2026]. We study graph misspecification, where the learner trusts one graph that differs

identification is already well developed [Anand et al., 2023, Ferreira and Assaad, 2025, Wahl et al., 2024].

Contributions.

- **Input contract.** We prove exact trajectory invariance for any optimizer that factors through the quotient (Theorem 1).
- **Value and refinement.** We characterize the coarsening gap, give sufficient conditions for zero gap, and connect a positive gap to incumbent regret.
- **Experiments.** Minimal SCMs isolate the two misspecification channels and the coarsening gap. A portability suite evaluates QCBO, QDCBO, and QMCBO against their base methods.

2 Background

A structural causal model $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, F, P(\mathbf{U}) \rangle$ induces observational and interventional distributions [Pearl, 2009]. Its latent projection is an acyclic directed mixed graph (ADMG) $\mathcal{G} = (\mathbf{V}, E)$ with directed causal edges and bidirected latent-confounding edges [Lee and Bareinboim, 2019].

Causal Bayesian optimization. Given manipulable variables $\mathbf{X} \subseteq \mathbf{V}$ and target Y , CBO [Aglietti et al., 2020] solves

$$(\mathbf{X}_s^*, \mathbf{x}_s^*) = \arg \min_{\substack{\mathbf{X}_s \in \mathbf{ES} \\ \mathbf{x}_s \in D(\mathbf{X}_s)}} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)], \quad (1)$$

where $D(\mathbf{X}_s)$ is the intervention domain. CBO maintains one Gaussian process (GP) per candidate intervention set. Its exploration set contains the minimal intervention sets (MIS) of the assumed graph [Lee and Bareinboim, 2018, Aglietti et al., 2020]. Writing $\mathcal{G}_{\bar{\mathbf{X}}_s}$ for the graph without incoming edges to $\mathbf{X}_s$ and $\text{An}(Y)$ for the ancestors of Y ,

$$\mathbf{ES} = \left\{ \mathbf{X}_s : \begin{array}{l} \emptyset \neq \mathbf{X}_s \subseteq \mathbf{X}, \\ \mathbf{X}_s \subseteq \text{An}(Y) \text{ in } \mathcal{G}_{\bar{\mathbf{X}}_s} \end{array} \right\}. \quad (2)$$

an arm is redundant when one of its variables ceases to be an ancestor of Y after intervention. Each GP uses a do-calculus estimate as prior mean, when available. Thus a structural hypothesis enters through

$$\mathcal{G} \mapsto (\mathbf{ES}(\mathcal{G}), \{m_s^{\mathcal{G}}\}_{s \in \mathbf{ES}(\mathcal{G})}) \mapsto \tau,$$

where τ is the optimization trajectory. A wrong graph can change a prior or the arm set itself.

Coarsening and C-DAGs.

Definition 1 (Coarsening [Madaleno et al., 2026]). *For a partition Π of $\mathbf{V}$, let $\chi : \mathbf{V} \rightarrow \Pi$ map each vertex to its cluster. Its directed quotient has*

$$E^\Pi = \{(\chi(u), \chi(v)) : (u, v) \in E, \chi(u) \neq \chi(v)\}.$$

Bidirected edges are mapped in the same way. The coarsening is valid when the directed quotient is acyclic.

$$Q_\Pi(\mathcal{G}) := (\Pi, \mathcal{G}^\Pi).$$

Define the coarse action family by

$$\mathcal{U}(\Pi) = \left\{ \bigcup_{C \in S} C : \emptyset \neq S \subseteq \Pi_M \right\},$$

where Π_M contains the manipulable clusters. For $A \subseteq \mathbf{X}$, define

$$\begin{aligned} \mu^*(A) &:= \min_{a \in D(A)} \mathbb{E}[Y \mid \text{do}(A = a)], \\ V(\Pi) &:= \min_{A \in \mathcal{U}(\Pi)} \mu^*(A), \\ \Delta(\Pi) &:= V(\Pi) - V(\Pi_\perp). \end{aligned}$$

and let $\Pi_\perp$ be the singleton partition. The MIS reduction in Lemma 1 connects this value to QCBO:

$$\min_{A \in \mathbf{ES}^\Pi} \mu^*(A) = \min_{A \in \mathcal{U}(\Pi)} \mu^*(A) = V(\Pi). \quad (3)$$

C-DAG identification is sound and complete over the fine graphs represented by the quotient [Anand et al., 2023]. Cluster-level Markov properties transfer, while faithfulness need not [Wahl et al., 2024].

Evaluation metrics. Let $y^* := V(\Pi_\perp)$ and let $\hat{a}_t$ be the incumbent after trial t . Incumbent mean regret is

$$r_t := \mu(\hat{a}_t) - y^* \geq 0, \quad R_T := \sum_{t=1}^T r_t.$$

The experiments also report paired trajectory differences between correct and misspecified inputs.

3 What coarsening buys

Write $\Pi' \preceq \Pi$ when every cluster of Π' is contained in a cluster of Π . An optimizer satisfies the quotient contract when its graph-dependent components use $\mathcal{G}$ only through $Q_\Pi(\mathcal{G})$.

Theorem 1 (Quotient invariance is method-agnostic). *If*

$$\mathcal{A}(\mathcal{G}, \mathcal{D}, \omega, \eta) = \bar{\mathcal{A}}(Q_\Pi(\mathcal{G}), \mathcal{D}, \omega, \eta),$$

then, for fixed data $\mathcal{D}$, random stream ω , and hyperparameters η ,

$$Q_\Pi(\mathcal{G}) = Q_\Pi(\mathcal{G}') \implies \tau_{\mathcal{A}}(\mathcal{G}) = \tau_{\mathcal{A}}(\mathcal{G}').$$

Thus within-cluster edits and quotient-redundant cross-cluster edits are protected. Quotient-visible edits are not.

3.1 What does protection cost?

Proposition 1 (Validity is not refinement-monotone).

$$\exists \Pi' \preceq \Pi : \quad \mathcal{G}^\Pi \text{ acyclic}, \quad \mathcal{G}^{\Pi'} \text{ cyclic}.$$

Validity must therefore be checked after refinement.

Proposition 2 (Refinement preserves retained arms). *Let $\Pi' \preceq \Pi$ be valid. If $A \in \mathbf{ES}^\Pi \cap \text{MIS}(\mathcal{G}^{\Pi'})$, then*

$$A \in \mathbf{ES}^{\Pi'}, \quad \phi_A^\Pi(P) = P(Y \mid \text{do}(A)).$$

Thus the coarse identification functional and the causal value remain valid under $\mathcal{G}^{\Pi'}$. MIS membership itself need not be monotone.

Proposition 3 (Monotonicity and price of coarsening). *For valid $\Pi' \preceq \Pi$,*

$$\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi'), \quad 0 \leq \Delta(\Pi') \leq \Delta(\Pi).$$

Moreover,

$$\Delta(\Pi) = 0 \iff \arg \min_{A \in \mathcal{U}(\Pi_\perp)} \mu^*(A) \cap \mathcal{U}(\Pi) \neq \emptyset.$$

Proposition 4 (Sufficient conditions for zero coarsening gap). *Assume a semi-Markovian SCM. For every $A \subseteq \mathbf{X}$, define*

$$\begin{aligned} \text{cl}_\Pi(A) &:= \bigcup \{C \in \Pi : C \cap A \neq \emptyset\}, \\ B &:= \text{cl}_\Pi(A) \setminus A. \end{aligned}$$

For each $a \in D(A)$ let B_a be the joint natural response under $\text{do}(A = a)$ and set

$$D_B(a) := \{b : (a, b) \in D(\text{cl}_\Pi(A))\}.$$

If, for all A, a ,

$$\text{supp}(B_a) \subseteq D_B(a), \quad B_a \perp Y_{a,b} \quad \forall b \in D_B(a),$$

then $V(\Pi) = V(\Pi_\perp)$ and $\Delta(\Pi) = 0$.

Corollary 1 (Graphical zero-gap condition). *In a semi-Markovian model, full intervention support and cluster-contained confounding imply the conditions of Prop. 4. Hence $\Delta(\Pi) = 0$. Here cluster-contained means that every backdoor path from a cluster member to Y meets another member of that cluster before reaching Y .*

Within this model class, the contrapositive is

$$\Delta(\Pi) > 0 \implies \begin{array}{c} \text{support failure or} \\ \text{confounding escaping a cluster.} \end{array}$$

Thus coarsening need not sacrifice the best attainable value. Even when $\Delta(\Pi) = 0$, it may change finite-sample regret, dimension, and intervention cost.

Refining the partition: Hierarchical QCBO (HQCBO).

Proposition 5 (Fixed-partition regret and refinement). *For a fixed partition,*

$$r_t \geq \Delta(\Pi), \quad R_T \geq T\Delta(\Pi).$$

If a valid refinement exposes an optimal arm and the base optimizer's incumbent is consistent on the finite arm set, then

$$r_t \rightarrow 0 \implies R_T/T \rightarrow 0.$$

The lower bound concerns the incumbent mean, not the exploratory action at trial t . HQCBO starts at Π and splits clusters according to a declared hierarchy when incumbent progress stalls. It rejects refinements whose quotient is cyclic, retains old-arm data, and initializes only new arms. The trigger and implementation are given in App. C.

Corollary 2 (Adaptive refinement invariance). *For visited partitions $\Pi_0, \dots, \Pi_J$,*

$$[\forall j, \mathcal{G}^{\Pi_j} = (\mathcal{G}')^{\Pi_j}] \implies \tau_H(\mathcal{G}) = \tau_H(\mathcal{G}').$$

Splitting a cluster exposes its internal edges in later phases. Edits inside clusters that are never split remain invisible.

4 Instantiations: QCBO, QDCBO, and QMCBO

Each construction replaces the base method's graph-coupled components by their quotient counterparts. Appendix C gives estimator and implementation details.

4.1 Arm-based: quotient CBO (QCBO)

QCBO receives a partition of the manipulable variables and its C-DAG. The target and non-manipulable variables remain singletons. Define $U_\Pi(S) := \bigcup_{C \in S} C$. Applying Eq. (2) to the quotient gives

$$\mathbf{ES}^\Pi = \left\{ U_\Pi(S) : \begin{array}{l} \emptyset \neq S \subseteq \Pi_M, \\ S \subseteq \text{An}(Y) \text{ in } \mathcal{G}_S^\Pi \end{array} \right\}. \quad (4)$$

Each arm sets one or more clusters jointly.

Identifiability and the two-tier prior. For each $A \in \mathbf{ES}^\Pi$, the complete ID algorithm [Shpitser and Pearl, 2006, Ananke] determines the prior:

$$m_A(a) = \begin{cases} \hat{\mathbb{E}}[Y \mid \text{do}(A = a)] & \text{if identified in } \mathcal{G}^\Pi, \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (5)$$

Non-identifiable arms learn from experimental data. Identification from the C-DAG is sound for every represented fine graph [Anand et al., 2023].

Assumption 1 (Estimator boundary). *The quotient is valid. The ID test is complete, while our numerical estimator is sound but not complete. Identifiable expressions outside its implemented class receive the uninformative prior.*

Proposition 6 (Identity recovery).

$$\begin{aligned} \Pi = \Pi_{\perp} &\implies \mathcal{G}^{\Pi} = \mathcal{G}^*, \\ \text{ES}^{\Pi} &= \text{MIS}(\mathcal{G}^*), \quad m_A^{\Pi} = m_A^{CBO}. \end{aligned}$$

4.2 Dynamic: quotient DCBO (QDCBO)

Dynamic CBO (DCBO) optimizes interventions across a dynamic Bayesian network [Aglietti et al., 2021]. QDCBO quotients each slice and its cross-slice transitions, then replaces emission and transition models by joint cluster mechanisms. Its exploration sets use Eq. (4) per slice. Information propagates to the next slice exactly as in DCBO.

Corollary 3 (QDCBO inherits the contract). *QDCBO satisfies Theorem 1 per slice. At $\Pi = \Pi_{\perp}$ it equals DCBO, and Prop. 3 applies to each slice’s reachable interventions.*

4.3 Model-based: quotient MCBO (QMCBO)

Model-based CBO (MCBO) composes one GP per causal mechanism and selects interventions using the Gaussian-process upper confidence bound acquisition function (GP-UCB) [Srinivas et al., 2010, Sussex et al., 2023]. QMCBO replaces the node mechanisms by

$$P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)}),$$

modeled as multi-output GPs with a coregionalization kernel [Bonilla et al., 2007]. These joint mechanisms are composed along the C-DAG.

Corollary 4 (QMCBO inherits the contract). *QMCBO satisfies Theorem 1. At $\Pi = \Pi_{\perp}$ it equals MCBO. Its reachable interventions are whole-cluster patterns, so Prop. 3 also applies.*

Proposition 7 (Soundness of cluster mechanisms). *Under causal sufficiency and valid Π , for every union of clusters S ,*

$$P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{C \notin S} P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})|_{\mathbf{x}_S = \mathbf{x}_S}.$$

The mechanism must be joint to retain within-cluster dependence.

Proposition 8 (Exposure: sub-cluster interventions are not identifiable from cluster mechanisms). *There exist causally sufficient SCMs M_1, M_2 , valid for the same Π , such that*

$$\begin{aligned} \mathcal{I}_{\Pi}(M_1) &= \mathcal{I}_{\Pi}(M_2), \\ P^{M_1}(Y \mid \text{do}(x_A)) &\neq P^{M_2}(Y \mid \text{do}(x_A)) \end{aligned}$$

for a strict subset $A \subsetneq C$. Here $\mathcal{I}_{\Pi}$ contains the joint cluster mechanisms and all whole-cluster interventional distributions.

5 Experiments

We first isolate quotient invariance, prior corruption, and the action-space gap on three SCMs with closed-form oracles. We then test all three optimizer families on their standard benchmarks. Full specifications are in App. B.

Protocol. For each minimal SCM we fix the true mechanisms, observational data ($n_{\text{obs}} = 100$), seeds, and evaluation budget. Only the assumed structure changes. We use 30 paired seeds, $T = 50$ trials, or 60 for MEDIATEDCHAIN, three initial points per arm, and unit costs. HQCBO uses $k = 5$ and $\delta = 10^{-3}$. The empirical cumulative-regret curves use the recorded best-so-far trajectory against the analytic y^* . They are descriptive and are not direct estimates of the mean-regret bound. Invariance is measured by $\sup_{s,t} |\Delta_{s,t}|$ between paired trajectories.

5.1 Experiment A: exploration-set corruption

PARALLEL PARENT uses the SCM from Fig. 1 with $\lambda = 1$. The fine MIS is $\{\{X_1\}, \{X_2\}, \{X_1, X_2\}\}$ and $y^* = 0$. Deleting $X_1 \rightarrow Y$ leaves only $\{X_2\}$, whose optimum is 4.25. Fine CBO therefore plateaus at the analytic gap (Fig. 2a). For $\Pi = \{\{X_1, X_2\}\}$ the deletion is quotient-invisible, and every QCBO trajectory is identical under the two graphs ($\sup |\Delta| = 0$). A spurious $X_1 \leftrightarrow Y$ is a quotient-visible negative control and changes the QCBO trajectory.

5.2 Experiment B: prior corruption without arm loss

FRONTDOOR uses $X_1 \rightarrow M \rightarrow Y$, latent $X_1 \leftrightarrow Y$, and a narrow-well objective whose optimum is $\text{do}(M = 1)$. Adding a spurious $X_1 \leftrightarrow M$ preserves the fine MIS but removes its identified priors. Fine CBO pays an empirical $\Delta R_{50} = 6.1 \pm 1.0$ and then reaches a similar final value (0.006 versus 0.001). Under $\Pi = \{\{X_1, M\}\}$ the edit is internal. QCBO is identical across conditions ($\sup |\Delta| = 0$).

Table 1: By Theorem 1, rows with an unchanged quotient are predicted invariant. The final column reports the measured paired deviation. PP = PARALLEL PARENT, FD = FRONT DOOR.

Cond.	Edit (assumed graph)	Quotient changed	Fine MIS changed	Fine prior changed	sup $ \Delta $
A1 (PP)	del $X_1 \rightarrow Y$ (quotient-redundant)	$\times$	$\checkmark$	—	0
A2 (PP)	add $X_1 \rightarrow X_2$ (intra-cluster bow)	$\times$	$\times$	$\checkmark$	0
A3 (PP)	assume $X_1 \leftrightarrow Y$ (quotient-visible)	$\checkmark$	$\times$	$\checkmark$	21.4
B1 (FD)	assume $X_1 \leftrightarrow M$ (intra-cluster conf.)	$\times$	$\times$	$\checkmark$	0

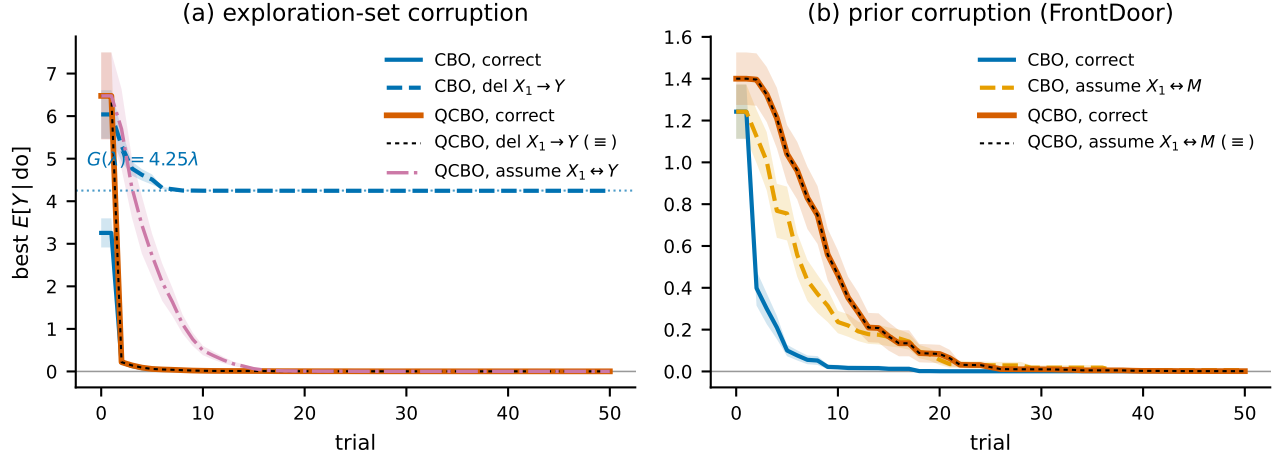


Figure 2: Correct and misspecified trajectories, mean $\pm$ s.e. over 30 seeds. (a) A wrong fine graph removes the optimal arm, while QCBO is unchanged for the quotient-invisible edit. (b) Prior corruption slows fine CBO transiently, while QCBO remains exactly invariant.

5.3 Experiment C: the price of coarsening, and buying it back

MEDIATEDCHAIN has $X_1 \rightarrow X_2 \rightarrow Y$, $X_2 = 2X_1 + \varepsilon_2$, $Y = (X_2 - 4)^2$, and $D(X_2) = [-2, 2]$. For $\Pi = \{\{X_1, X_2\}\}$,

$$V(\Pi) = 4, \quad y^* = 0.04, \quad \Delta(\Pi) = 3.96.$$

The support condition in Prop. 4 fails because $\text{do}(X_1 = 2)$ lets X_2 respond near 4, outside its intervention box. CBO, QCBO, and HQCBO finish at 0.040, 4.000, and 0.040. HQCBO refines at trial 7.5 ± 0.1 . The fixed-partition empirical regret grows linearly, while the refined curve flattens after the fine arm appears.

5.4 Experiment D: portability across families

We compare each quotient method with its base in the base paper’s released stack and metric. The suite contains three CBO tasks (10 seeds), three DCBO tasks (20 seeds), and two MCBO tasks (5 seeds). Graphs and full protocols are in Apps. D and E.

Table 2: Realized price of coarsening per family, each pair on the base method’s own experiment suite (final Y , mean $\pm$ s.e.). Price is signed so that positive means the quotient did worse in the task’s direction.

Family	Experiment	Base final	Quotient final	Price	Task
CBO vs QCBO	Toy	-2.16 ± 0.00	-2.13 ± 0.03	+0.03	min
CBO vs QCBO	Synthetic	-3.14 ± 0.19	-1.25 ± 0.07	+1.89	min
CBO vs QCBO	Coral	36.08 ± 0.02	36.42 ± 0.01	+0.34	min
DCBO vs QDCBO	stat (T=3)	-6.14 ± 0.05	-6.43 ± 0.02	-0.29	min
DCBO vs QDCBO	ind (T=3)	-3.12 ± 0.06	-5.57 ± 0.15	-2.45	min
DCBO vs QDCBO	nonstat (T=3)	8.03 ± 0.90	6.33 ± 0.29	-1.69	min
MCBO vs QMCBO	ToyGraph	1.14 ± 0.27	2.15 ± 0.01	-1.01	max
MCBO vs QMCBO	PSAGraph	-5.15 ± 0.00	-5.15 ± 0.00	-0.00	max

Results. Table 2 carries the final values. The quotient matches or beats its base on six of eight tasks. On ToyGraph, QMCBO reaches 2.150 ± 0.013 while MCBO reaches 1.143 ± 0.271 . On PSAGraph both finish at -5.15 ± 0.00 , with QMCBO using $3.0\times$ less wall-clock time. Synthetic has the largest positive gap because its optimum splits a cluster. Under quotient-invisible perturbations, QMCBO is identical on all 10 environment-seed pairs, while MCBO changes on 8 of 10. QDCBO is likewise invariant to its intra-slice edit.

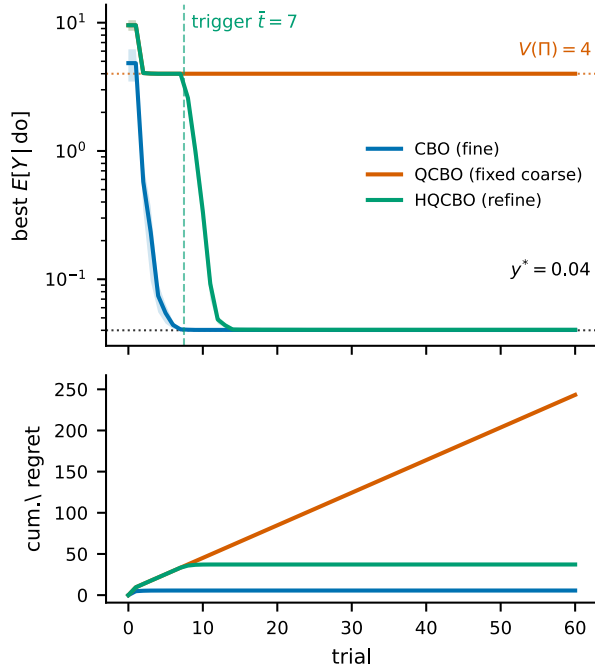


Figure 3: MEDIATEDCHAIN over 30 seeds. Top: mean $\pm$ s.e. best-so-far value and mean refinement time. Bottom: mean empirical cumulative regret, excluding the shared initial incumbent. Refinement exposes the fine optimal arm.

6 Related work

QCBO neither assumes a full graph nor learns one. GACBO maintains a DAG posterior [Mukherjee et al., 2024], CEO uses graph entropy [Branchini et al., 2023], and Durand et al. [2025] identify $\text{Pa}(Y)$ under a Y -leaf assumption. Other work assumes an equivalence class [Park et al.]. We instead use C-DAG identification [Anand et al., 2023, Ferreira and Assaad, 2025] as a fixed, coarser input. Causal abstraction has also been used to coarsen the outcome as a low-fidelity proxy [Zeitler, 2025]. We coarsen manipulable inputs.

7 Discussion and limitations

The guarantee is behavioral and requires a correct quotient. Faithfulness need not transfer to clusters [Wahl et al., 2024]. The experiments use unit intervention costs. Constructing the exploration set requires at most $2^K - 1$ MIS and ID checks for K manipulable clusters. QMCBO, like MCBO, assumes causal sufficiency. A wrong prior can be overcome if its arm remains available, while a removed or unavailable arm cannot be recovered without changing the action space. Learning the hierarchy and controlling unnecessary-refinement cost remain open.

8 Conclusion

Quotient inputs make CBO invariant to structural errors they do not represent. Under the support and confounding conditions of Prop. 4, this robustness has zero action-space gap. Otherwise refinement can recover fine interventions. The same contract applies to arm-based, dynamic, and model-based CBO.

References

- V. Aglietti, X. Lu, A. Paleyes, and J. González. Causal Bayesian Optimization, May 2020.
- V. Aglietti, N. Dhir, J. González, and T. Damoulas. Dynamic Causal Bayesian Optimization. In *Advances in Neural Information Processing Systems*, volume 34, pages 10549–10560. Curran Associates, Inc., 2021.
- V. Aglietti, A. Malek, I. Ktena, and S. Chiappa. Constrained Causal Bayesian Optimization. In *Proceedings of the 40th International Conference on Machine Learning*, pages 304–321. PMLR, July 2023.
- T. V. Anand, A. H. Ribeiro, J. Tian, and E. Bareinboim. Causal effect identification in cluster DAGs. <https://dl.acm.org/doi/10.1609/aaai.v37i10.26435>, 2023.
- Ananke. Ananke: A module for causal inference — ananke-causal documentation.
- Anonymous. Causal Bayesian optimization: Foundations, methods, and applications. <https://openreview.net/forum?id=XT6DC37m5I>, 2026. Under review, Transactions on Machine Learning Research.
- E. Bonilla, K. Chai, and C. Williams. Multi-task Gaussian Process Prediction. In *Advances in Neural Information Processing Systems*, volume 20. Curran Associates, Inc., 2007.
- N. Branchini, V. Aglietti, N. Dhir, and T. Damoulas. Causal Entropy Optimization. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, pages 8586–8605. PMLR, Apr. 2023.
- J. Durand, Y. Annadani, S. Bauer, and S. Parbhoo. Causal Bayesian Optimization with Unknown Graphs, Mar. 2025.
- S. Ferreira and C. K. Assaad. Identifying Macro Causal Effects in C-DMGs over DMGs. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, Oct. 2025.
- R. Garnett. *Bayesian Optimization*. Cambridge University Press, Cambridge, 2023. doi: 10.1017/9781108348973.

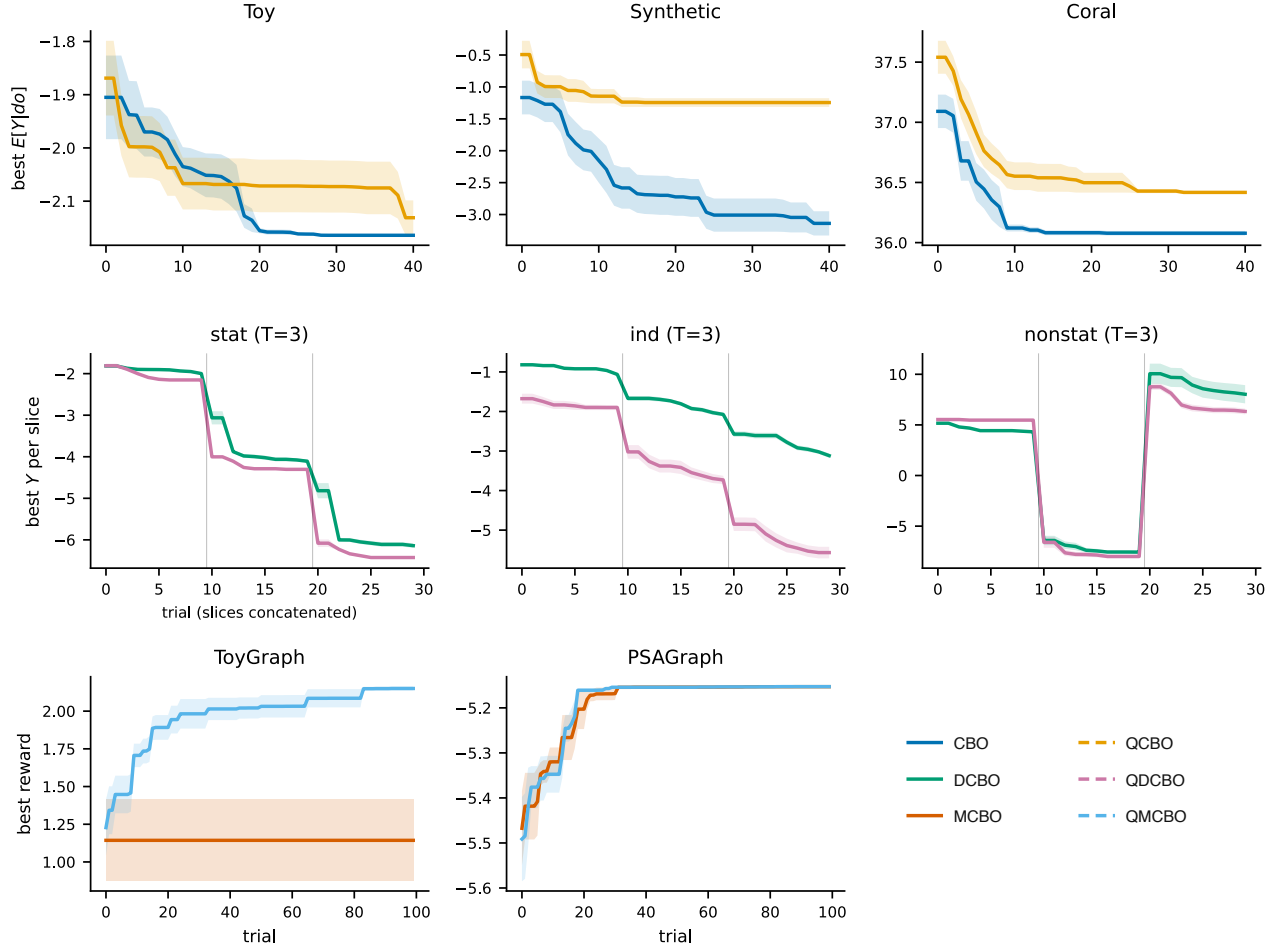


Figure 4: Quotient methods and their bases on the base papers’ benchmarks, mean $\pm$ s.e. Top: minimization for CBO. Middle: DCBO slices. Bottom: maximization for MCBO.

- S. Lee and E. Bareinboim. Structural Causal Bandits: Where to Intervene? *Advances in Neural Information Processing Systems*, 2018.
- S. Lee and E. Bareinboim. Structural Causal Bandits with Non-Manipulable Variables. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01): 4164–4172, July 2019. ISSN 2374-3468. doi: 10.1609/aaai.v33i01.33014164.
- F. Madaleno, P. Misra, and A. Markham. Coarsening Causal DAG Models: Conference on Causal Learning and Reasoning (CLearR). *Proceedings of the Fifth Conference on Causal Learning and Reasoning*, 2026.
- S. Mukherjee, M. Zhang, S. Flaxman, and S. J. Vollmer. Graph Agnostic Causal Bayesian Optimization, Nov. 2024.
- M. W. Park, A. Arditì, E. Bareinboim, and S. Lee. Structural Causal Bandits under Markov Equivalence.
- J. Pearl. *Causality*. Cambridge University Press, Cambridge, 2 edition, 2009. doi: 10.1017/CBO9780511803161.
- I. Shpitser and J. Pearl. Identification of Joint Interventional Distributions in Recursive Semi-Markovian Causal Models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2006.
- N. Srinivas, A. Krause, S. M. Kakade, and M. Seeger. Gaussian process optimization in the bandit setting: No regret and experimental design. In *Proceedings of the 27th International Conference on Machine Learning*, pages 1015–1022. Omnipress, 2010.
- S. Sussex, A. Makarova, and A. Krause. Model-Based Causal Bayesian Optimization. 2023.
- J. Wahl, U. Ninad, and J. Runge. Foundations of causal discovery on groups of variables. *Journal of Causal Inference*, 12(1):20230041, July 2024. ISSN 2193-3685. doi: 10.1515/jci-2023-0041.
- J. Zeitler. Towards MFACBO: Multi-Fidelity Abstrac-

tion Causal Bayesian Optimization in the Context of the Abstraction-Fidelity Connection. In *UAI 2025 Workshop on Causal Abstractions and Representations*, Aug. 2025.

A Proofs

Notation is as in §4.1 and §3.

Proof of Theorem 1. Write the run as decisions $d_1, d_2, \dots$ and datasets $\mathcal{D}_0, \mathcal{D}_1, \dots$, with

$$d_t = f_t(Q_\Pi(\mathcal{G}), \mathcal{D}_{t-1}, \omega_t, \eta).$$

When $Q_\Pi(\mathcal{G}) = Q_\Pi(\mathcal{G}')$, the first arguments agree. The fixed-objective protocol also fixes $\mathcal{D}_0$, the random stream, and the hyperparameters. Thus d_1 and the resulting $\mathcal{D}_1$ agree. Induction gives equality of every decision, dataset, and stopping event, hence of the full trajectories. Within-cluster edits and cross-cluster edits redundant in the quotient leave $Q_\Pi(\mathcal{G})$ unchanged, which proves the stated scope. $\square$

Proof of Prop. 7. Under causal sufficiency the fine SCM is Markov relative to $\mathcal{G}$ with mutually independent exogenous noises, so for any intervention $\text{do}(\mathbf{x}_S)$ on a union of clusters S the fine truncated factorization gives $P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{k \notin S} P(X_k \mid \mathbf{X}_{\text{pa}_G(k)}) \big|_{\mathbf{x}_S = \mathbf{x}_S}$. Group the factors by the clusters of Π . For a cluster $C \not\subseteq S$, the product of its members' factors is, by the chain rule applied in any topological order of $\mathcal{G}$ restricted to C , exactly the joint conditional $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$, where $\text{pa}(C)$, the union of C 's parent clusters, contains every out-of-cluster variable appearing in a member's conditioning set (validity of Π guarantees the quotient is acyclic, so $\text{pa}(C)$ is well defined and precedes C in C-DAG order). The fine factorization therefore equals the product over clusters of the joint cluster mechanisms with the clusters of S clamped at $\mathbf{x}_S$: the cluster-level truncated factorization. Note both uses of sufficiency: independent noises across clusters (no bidirected edges) and the absence of unobserved common causes that would make a member's conditional depend on variables outside $\text{pa}(C) \cup C$. Replacing $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$ by the product of its per-member marginals would be exact iff the members are conditionally independent given $\text{pa}(C)$, in particular when C contains no intra-cluster edges. This proves why the construction uses a joint mechanism. $\square$

Proof of Prop. 8. Take $\mathbf{X} = \{X_2, X_3\}$ as one root cluster C , target $Y = X_2 + \varepsilon_Y$ its child, all noises standard normal and independent (sufficiency holds, and any Π clustering $\{X_2, X_3\}$ is valid). Let M_1 have $X_2 = \varepsilon_2$ and $X_3 = aX_2 + \varepsilon_3$. Let M_2 have $X_3 = \varepsilon'_3$, $X_2 = a'X_3 + \varepsilon'_2$, with $a' = a/(1+a^2)$, $\text{Var}(\varepsilon'_3) = 1+a^2$, and $\text{Var}(\varepsilon'_2) = 1-a^2/(1+a^2)$. Both induce the same zero-mean Gaussian joint on (X_2, X_3) , with covariance $\begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix}$, hence the same joint cluster mechanism (here, the same root-cluster distribution), and the same $P(Y \mid \text{do}(\mathbf{x}_C))$ for the whole-cluster intervention (both set X_2 directly). But under the sub-cluster intervention $\text{do}(X_3 = x)$: in M_1 , $X_2 = \varepsilon_2$ is unaffected, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = 0$ for all x . In M_2 , $X_2 = a'x$, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = a'x \neq 0$ for $x \neq 0$, $a \neq 0$. Every quotient-respecting input (the C-DAG, the joint cluster mechanisms, all whole-cluster interventional distributions) coincides between M_1 and M_2 , yet the sub-cluster query differs. No function of those inputs can answer it. $\square$

Proof of Prop. 6. Singleton clusters make the quotient map the identity on $\mathcal{G}^*$, so $\mathcal{G}^\Pi = \mathcal{G}^*$, unions of singleton clusters are exactly the subsets of $\mathbf{X}$, and the MIS condition of Eq. (4) coincides verbatim with Eq. (2). The prior construction poses the same do-calculus queries CBO poses on the full graph, so priors coincide arm-for-arm. $\square$

Lemma 1 (MIS reduction). *Let G be an ADMG with manipulable vertices M and target Y , let $\emptyset \neq \mathbf{W} \subseteq M$, and set $\mathbf{W}' = \mathbf{W} \cap \text{An}(Y)$ computed in $G_{\bar{\mathbf{W}}}$. Then (i) $P(Y \mid \text{do}(\mathbf{W} = \mathbf{w})) = P(Y \mid \text{do}(\mathbf{W}' = \mathbf{w}|_{\mathbf{W}'}))$ for every $\mathbf{w}$, (ii) $\mathbf{W}'$ is a MIS (or empty, when no member reaches Y), and (iii) consequently $\min_{\mathbf{x}_s \in \text{MIS}(G)} \mu^*(\mathbf{X}_s) = \min_{\emptyset \neq \mathbf{W} \subseteq M} \mu^*(\mathbf{W})$: restricting the search to MIS loses no value.*

Proof. (i) In $G_{\bar{\mathbf{W}}}$ the members of $\mathbf{W} \setminus \mathbf{W}'$ have no directed path to Y , so Y is unaffected by their values under the intervention. Dropping them from the intervention changes no mechanism on any directed path into Y (do-calculus rule 3). (ii) A directed path from $w' \in \mathbf{W}'$ to Y in $G_{\bar{\mathbf{W}}}$ passes through no intervened vertex (their incoming edges are cut), hence through no member of $\mathbf{W} \supseteq \mathbf{W}'$. The same path exists in $G_{\bar{\mathbf{W}'}}$, which only restores edges, so $\mathbf{W}' \subseteq \text{An}(Y)$ there. (iii) By (i) and (ii) every candidate's value is attained by a MIS, and $\text{MIS}(G)$ is a subset of the candidates. $\square$

Proof of Prop. 1. Take $\mathbf{X} = \{a_1, a_2, b_1, b_2\}$ with fine edges $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$. The coarsest partition $\Pi = \{\{a_1, a_2, b_1, b_2\}\}$ is valid: both edges are intra-cluster, so the quotient has a single manipulable node and is

acyclic. Its refinement $\Pi' = \{\{a_1, a_2\}, \{b_1, b_2\}\}$ exposes both edges as inter-cluster ($a_1 \rightarrow b_1$ contributes $A \rightarrow B$ and $b_2 \rightarrow a_2$ contributes $B \rightarrow A$), inducing the 2-cycle $A \rightleftarrows B$, so $\mathcal{G}^{\Pi'}$ is cyclic and Π' is invalid. Decidability is immediate from the acyclicity clause of Def. 1. $\square$

Proof of Prop. 2. (i) Membership at Π' is the stated hypothesis (any union of Π -clusters is a union of Π' -clusters, so the candidate exists at Π' , and being a MIS of $\mathcal{G}^{\Pi'}$ puts it in $\mathbf{ES}^{\Pi'}$). For the identification functional, quotienting is compositional. The graph $\mathcal{G}^{\Pi}$ is the quotient of $\mathcal{G}^{\Pi'}$ under the induced partition of Π' -clusters, so every fine graph compatible with $\mathcal{G}^{\Pi'}$ is compatible with $\mathcal{G}^{\Pi}$. Identifiability from $\mathcal{G}^{\Pi}$ means a single functional of the observational distribution equals $\mathbb{E}[Y \mid \text{do}(\mathbf{X}_s)]$ across all SCMs whose graph is compatible with $\mathcal{G}^{\Pi}$ [Anand et al., 2023]. A fortiori the same functional works across the smaller class compatible with $\mathcal{G}^{\Pi'}$. MIS membership can fail after refinement because restoring distinctions between cluster members may make one member redundant under intervention. $\square$

Proof of Prop. 3. By Lemma 1(iii) applied to $\mathcal{G}^{\Pi}$, $V(\Pi) = \min_{\mathbf{X}_s \in \mathbf{ES}^{\Pi}} \mu^*(\mathbf{X}_s)$ equals the minimum of μ^* over all non-empty unions of Π -clusters (the value μ^* of a cluster union is a property of the underlying SCM, the same at every partition that can express the union). Unions of Π -clusters are unions of Π' -clusters and, at $\Pi_{\perp}$, all non-empty subsets of $\mathbf{X}$, so the three minima are over nested candidate families: $V(\Pi_{\perp}) \leq V(\Pi') \leq V(\Pi)$. Equality $V(\Pi) = V(\Pi_{\perp})$ holds iff the unrestricted minimum is attained by a union of Π -clusters, and by Lemma 1(iii) at $\Pi_{\perp}$ the unrestricted minimum is min over Eq. (1)'s exploration set. This is exactly the stated condition. $\square$

Proof of Prop. 5. By Lemma 1(iii), $y^* = V(\Pi_{\perp})$. A fixed partition executes only arms in $\mathcal{U}(\Pi)$, so every incumbent satisfies

$$r_t = \mu(\hat{a}_t) - y^* \geq V(\Pi) - V(\Pi_{\perp}) = \Delta(\Pi).$$

Summation gives $R_T \geq T\Delta(\Pi)$. If a visited refinement has $\Delta = 0$, its exploration set contains an optimal arm by Prop. 3. Incumbent consistency gives $r_t \rightarrow 0$, and Cesàro averaging gives $R_T/T \rightarrow 0$. $\square$

Proof of Prop. 4. Fix A and a . Consistency and the assumed cross-world independence give

$$\begin{aligned} \mathbb{E}[Y_a] &= \int \mathbb{E}[Y_{a,b} \mid B_a = b] dP(B_a = b) \\ &= \int \mathbb{E}[Y_{a,b}] dP(B_a = b) \\ &\geq \inf_{b \in D_{B(a)}} \mathbb{E}[Y_{a,b}], \end{aligned}$$

where the support inclusion justifies the last line. Therefore

$$\mu^*(\text{cl}_{\Pi}(A)) \leq \mu^*(A).$$

The closure of every fine arm is a union of clusters, so $V(\Pi) \leq V(\Pi_{\perp})$. Conversely, $\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi_{\perp})$ implies $V(\Pi) \geq V(\Pi_{\perp})$. Hence equality holds. $\square$

Proof of Cor. 1. Represent the semi-Markovian SCM using mutually independent exogenous variables, with each latent common cause represented by one shared exogenous variable. Full intervention support gives $\text{supp}(B_a) \subseteq D_{B(a)}$.

Let $\mathbf{U}_B$ contain the exogenous variables that determine B_a , and let $\mathbf{U}_Y$ contain those that determine $Y_{a,b}$ after $\text{cl}_{\Pi}(A)$ is clamped. If some $U^* \in \mathbf{U}_B \cap \mathbf{U}_Y$, then U^* has a directed path to a member of B and another to Y that avoids the clamped closure. Taking the first closure member on the former path yields a backdoor path from a cluster member to Y whose interior avoids the rest of its cluster. This contradicts cluster-contained confounding. Hence $\mathbf{U}_B \cap \mathbf{U}_Y = \emptyset$. Mutual independence of the exogenous variables gives $B_a \perp Y_{a,b}$, so Prop. 4 applies.

Both premises can be necessary. In MEDIATEDCHAIN, setting $X_1 = 2$ lets the natural response of X_2 reach 4, while hard interventions constrain X_2 to $[-2, 2]$. The support condition fails and $\Delta(\Pi) = 3.96$. For an independence counterexample, let $C = \{A, B\}$, $B = U$, and $Y = -BU + (A - 1)^2$ with $U \sim \mathcal{N}(0, 1)$. Then $\mathbb{E}[Y \mid \text{do}(A = 1)] = -1$, whereas $\mathbb{E}[Y \mid \text{do}(A = a, B = b)] = (a - 1)^2 \geq 0$. Whole-cluster intervention severs the useful dependence, giving $\Delta(\Pi) = 1$ despite unrestricted domains. $\square$

Proof of Cor. 2. Induct over phases. Within phase j , every graph-dependent component factors through $(\Pi_j, \mathcal{G}^{\Pi_j})$, while the trigger and split are functions of the trajectory and declared hierarchy. Equal phase- j quotients and prefixes therefore give the same phase trajectory by Theorem 1, followed by the same trigger and partition Π_{j+1} . Repeating through Π_J proves equality of the adaptive trajectories. An edit inside a never-split cluster remains invisible at every visited quotient. An edit inside a split cluster is invisible before the split and may become visible afterward. $\square$

B MinimalBench: specification and oracles

The three SCMs share one parameterization: Gaussian roots, manipulable domains $[-3, 3]$ (except the MEDIATEDCHAIN policy box), unit intervention costs, target noise $\sigma_Y = 0.1$, and no clipping. Every arm value is closed-form. The structural module that defines them (`ccbo/minibench.py`) is the same one the methods, the perturbation registry, the structural tests, and Table 1’s emitter consume, so specification and experiment cannot drift. Observational datasets (5,000 rows, seed 0, with every run using the first $n_{\text{obs}}=100$) are produced by released generators whose built-in oracle gates assert the Monte-Carlo arm values against the closed forms below (tolerance 0.05) at build time.

PARALLEL PARENT (Experiment A). Latent variable U , manipulable variables X_1, X_2 , and target Y :

$$U = 0.4\epsilon_0, \quad X_i = U + 0.3\epsilon_i, \quad Y = \lambda(X_1 - 2)^2 + (X_2 + 2)^2 + \sigma_Y\epsilon_Y,$$

with $\lambda = 1$. The latent projection carries $X_1 \leftrightarrow X_2$ (corr = 0.64), and $\text{Var}X_i = 0.25$. Coarse partition $C_1 = \{X_1, X_2\}$. Closed forms: $\mathbb{E}[Y \mid \text{do}(x_1, x_2)] = \lambda(x_1 - 2)^2 + (x_2 + 2)^2$, so $y^* = 0$ at the joint arm. Also, $\mathbb{E}[Y \mid \text{do}(x_1)] = \lambda(x_1 - 2)^2 + 4.25$ (the free co-parent’s bowl plus its variance) and $\mathbb{E}[Y \mid \text{do}(x_2)] = 4.25\lambda + (x_2 + 2)^2$. Under A1 (delete $X_1 \rightarrow Y$) the fine MIS is $\{X_2\}$ alone. Since $X_1 \notin \text{An}(Y)$ in every mutilation, the best reachable fine value is $G(\lambda) = \lambda(4 + 0.25) = 4.25\lambda$, the analytic gap of §5.1. Condition A1 deletes $X_1 \rightarrow Y$ and is quotient-redundant. Condition A2 adds $X_1 \rightarrow X_2$, creating an intra-cluster bow with latent $X_1 \leftrightarrow X_2$. The $\text{do}(X_1)$ effect loses identification, but membership is intact. Condition A3 assumes $X_1 \leftrightarrow Y$ (quotient-visible: the C-DAG gains $C_1 \leftrightarrow Y$ and the cluster arm drops to the uninformative tier, and QCBO keeps running).

FRONT DOOR (Experiment B). Latent variable U , manipulable variables X_1, M , and target Y :

$$U = 0.4\epsilon_0, \quad X_1 = U + 0.3\epsilon_1, \quad M = 2X_1 + 0.2\epsilon_M, \\ Y = 2(1 - e^{-(M-1)^2/(2 \cdot 0.3^2)}) + U + \sigma_Y\epsilon_Y.$$

The latent projection carries $X_1 \leftrightarrow Y$. The target is a narrow well (width 0.3 over the 6-wide domain): prior corruption is a global-search phenomenon, and on a smooth quadratic the same corruption measurably costs nothing (§5.2). Fine MIS = $\{X_1\}, \{M\}$ (chain). The effect of $\text{do}(M)$ is identified by backdoor through X_1 , and $\text{do}(X_1)$ by the front-door functional through M . Gaussian smoothing gives the closed forms: $\mathbb{E}[Y \mid \text{do}(m)] = 2(1 - e^{-(m-1)^2/0.18})$, so $y^* = 0$ at $\text{do}(M=1)$. Moreover, $\mathbb{E}[Y \mid \text{do}(x_1)] = 2(1 - \frac{0.3}{\sqrt{0.3^2 + 0.2^2}} e^{-(2x_1-1)^2/(2(0.3^2+0.2^2))})$, best ≈ 0.336 at $x_1 = 1/2$. The observational mean is $\mathbb{E}[Y] \approx 1.637$. The coarse partition $\{X_1, M\}$ quotients to the bow $C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$. The cluster arm runs on the uninformative tier in *every* condition, and setting M directly makes the partition lossless ($V(\Pi) = y^* = 0$). Condition B1 assumes a spurious intra-cluster confounder $X_1 \leftrightarrow M$: both fine arms lose identification (the backdoor set $\{X_1\}$ for $\text{do}(M)$ is invalidated by the induced collider path, the front-door premise for $\text{do}(X_1)$ by the confounded $X_1 \rightarrow M$ edge), while the quotient drops intra-cluster bidirected edges and is unchanged.

MEDIATED CHAIN (Experiment C). There are no latent variables. The manipulable variables are X_1, X_2 and the target is Y :

$$X_1 = 0.5\epsilon_1, \quad X_2 = 2X_1 + 0.2\epsilon_2, \quad Y = (X_2 - 4)^2 + \sigma_Y\epsilon_Y,$$

with $D(X_1) = [-3, 3]$ and the *policy box* $D(X_2) = [-2, 2]$. Fine MIS = $\{X_1\}, \{X_2\}$. The joint is not minimal on a chain because mutilating X_2 severs X_1 from $\text{An}(Y)$ (Prop. 2), so the joint intervention exists only as the quotient’s cluster arm, which must clamp X_2 into the box. Closed forms: $\mathbb{E}[Y \mid \text{do}(x_1)] = (2x_1 - 4)^2 + 0.04$, so $y^* = 0.04$ at $\text{do}(X_1=2)$, which drives X_2 ’s natural response to 4, outside the box. We have $\mathbb{E}[Y \mid \text{do}(x_2)] = (x_2 - 4)^2$,

whose best value over the box is 4. Hence $V(\Pi) = 4$ and $V(\Pi) - y^* = 3.96$ (Prop. 3). The support premise of Prop. 4 fails by construction). The declared hierarchy splits $C_1 = \{X_1, X_2\}$ into singletons. One split reaches the identity partition, whose MIS contains the optimal $\text{do}(X_1)$ arm.

Structural tests and oracle gates. Dataset-free structural tests assert, per condition, the exact fine MIS, the C-DAG signature (in)equalities, and the prior-tier flips claimed above, and end-to-end regressions assert QCBO’s trajectory invariance (arm choices and intervention values exactly, with incumbents matching to 10^{-8}) under every protected condition and its divergence under A3. The data generators refuse to write datasets whose Monte-Carlo arm values, correlations, or policy-box geometry contradict the closed forms.

C Fixed-objective protocol: per-method details

The invariant is that every method optimizes the *same* objective, the true SCM’s noise-free `intervention_function`, and only the *structure each method reasons with* is perturbed. We realize this per method as follows.

QCBO (finest and coarse). The objective is read from the original graph’s SEM, while the structure is supplied separately through the assumed graph name (`CoarsenedGraph(..., assumed_graph_name=variant)`). Swapping the assumed graph changes the quotient, the exploration set, the identification formulas, and the prior, but never the SEM that generates outcomes. *Prior pipeline.* An identifiable arm’s prior mean instantiates the identified formula as a pipeline of GP-based plug-ins (backdoor $\rightarrow$ front-door $\rightarrow$ g -computation) on the bidirected-edge expansion of $\mathcal{G}^\Pi$. A non-identifiable arm, or an identifiable query the sound-but-incomplete pipeline cannot instantiate, shares the common uninformative prior, the observational mean of Y . Because clusters are manipulable-only and interventions assign full clusters, the cluster-level formulas expand exactly into fine-grained regressions on observational data.

CBO (full-DAG identification). CBO is QCBO’s backend at the identity partition (Prop. 6): it runs the same complete identification algorithm, only on the full DAG rather than a quotient. The injection is the same assumed-graph swap as for QCBO (the perturbed structure reaches the exploration set and the identification formulas that build the prior), while outcomes are always drawn from the true SEM. Its trajectory is identical to QCBO’s at the finest partition on every graph.

Exploration-set disclosure. Both methods use the MIS of the assumed graph (Eq. 2), computed by the released code and verified against hand-derived Lee–Bareinboim ground truth per benchmark (for the minimal suite, the per-condition sets are also asserted by the structural tests of App. B). The CBO authors’ shipped code deviates from that definition in two places, which we correct for both methods in the family suite: on `ToyGraph` it lists the non-minimal arm $\{X, Z\}$ (mutilating Z cuts $X \rightarrow Z$, so $\text{do}(X, Z) \equiv \text{do}(Z)$ and the joint arm is redundant), and on `CoralGraph` it truncates the list at subsets of size three (an ad-hoc cap; the definition yields all 31 non-empty subsets, as every manipulable variable keeps a directed path to Y through never-intervened non-manipulables under any mutilation).

HQCBO. Phase 1 is the QCBO configuration run by the same code path with the same seed (the trigger is a function of the coarse trajectory alone), so its pre-trigger prefix is byte-identical to the QCBO run, which we assert per seed. The refined phase re-enters the same backend with the split partition, the unioned exploration set, and the carried per-arm data (§3.1); the vendored loop forces an observation step at the start of a phase, a small bias *against* refinement that we keep.

Objective-identity verification. For all methods the decoupling holds by construction: outcomes are always evaluated on the true graph’s SEM, and the perturbation enters only through the separately-supplied assumed-graph variant (including assumed latents, as in A3 and B1). The released data generators additionally refuse to build any dataset whose Monte-Carlo arm values contradict the closed-form oracles (App. B): feeding a method a wrong graph changes only its causal reasoning, never the problem it is solving.

D Dataset graphs and their quotients

For every dataset in the paper, the figures below show the fine DAG with its coarse partition and the quotient C-DAG the method actually consumes. They are regenerated from the released structure registry by



Figure 5: **ParallelParent (MinimalBench, Exp. A)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The intra-cluster $X_1 \leftrightarrow X_2$ disappears in the quotient; deleting $X_1 \rightarrow Y$ leaves $C_1 \rightarrow Y$ alive through $X_2 \rightarrow Y$ (quotient-redundant).



Figure 6: **FrontDoor (MinimalBench, Exp. B)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): the cluster arm runs on the uninformative prior tier in every condition, so the assumed intra-cluster $X_1 \leftrightarrow M$ cannot change it.

`scripts/emit_dataset_tikz.py`, so the pictures cannot drift from the code. On **PARALLELParent**, note how the intra-cluster latent $X_1 \leftrightarrow X_2$ disappears in the quotient while $C_1 \rightarrow Y$ survives any single parent-edge deletion. This is the redundancy behind Experiment A; the **FRONTDOOR** and **ToyGraph** pairs both quotient to the same cluster bow, illustrating Prop. 8.

E Implementation and reproducibility

All methods share one surrogate/acquisition backend (the vendored CBO loop), so comparisons differ only in the assumed structure. **Surrogate**. Each exploration-set arm carries a `GPY GPRegression` model with an RBF kernel (signal variance and lengthscale initialized to 1, ARD off) and observation noise fixed near zero (10^{-10} , matching the deterministic mechanisms). The causal prior supplies the do-calculus estimate as the GP *mean function*, so Δ -metrics isolate the effect of a wrong prior/exploration set from surrogate noise. **Acquisition**. Causal expected improvement [Garnett, 2023], maximized over the box-constrained intervention domain by gradient-based L-BFGS. **Protocol (minimal suite)**. Unit per-variable intervention costs, 3 initial interventional points per arm, $T=50$ trials (60 on **MEDIATEDCHAIN**), 30 seeds, $n_{\text{obs}}=100$ from one shared dataset per SCM; the family suite follows each base paper’s own protocol (§5.4). Every run additionally logs the selected arm and intervention values per trial, so invariance is checked on *decisions*, not only on incumbent values. **Invariance tolerance**. Arm-choice and intervention-value sequences are compared exactly; incumbent trajectories at 10^{-8} (GP-solver floating point). On every protected condition of Table 1 the sequences are identical and $\sup |\Delta| = 0$; the quotient-visible control moves trajectories on every seed. **Artifacts**. The **MinimalBench** structural module (`ccbo/minibench.py`), its data generators with built-in oracle gates, the dataset-free structural tests and end-to-end invariance regressions, the suite runner (`scripts/run_minimal_suite.py`), the native two-phase HQCBO (`ccbo/minimal_suite.py`), and every table/figure emitter are released, so all reported numbers regenerate from the released SCMs and seeds.



Figure 7: **MediatedChain (MinimalBench, Exp. C)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The policy box restricts $D(X_2)$ to $[-2, 2]$; the fine optimum $\text{do}(X_1=2)$ drives X_2 to 4, outside the box.



Figure 8: **ToyGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): $\text{do}(C_1)$ is not identifiable from the C-DAG, so the arm runs on the uninformative prior tier.

F Full results

Table 4 positions QCBO in the method landscape (§6). Tables 3 and 5 give the per-environment detail behind §5.4’s model-based rows: finals and wall-clock on MCBO’s own tasks, and the within-family invariance counts under the intra-cluster perturbation of the model’s graph view. Per-seed CSVs for every minimal-suite unit (trial, incumbent, cumulative cost, selected arm, and intervention values) regenerate from the released runners (App. E).

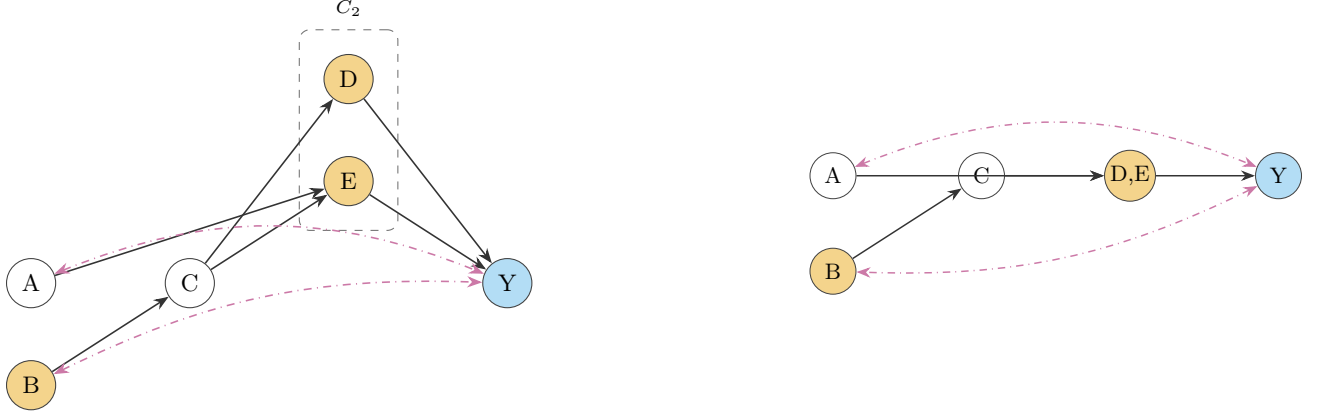


Figure 9: **CompleteGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

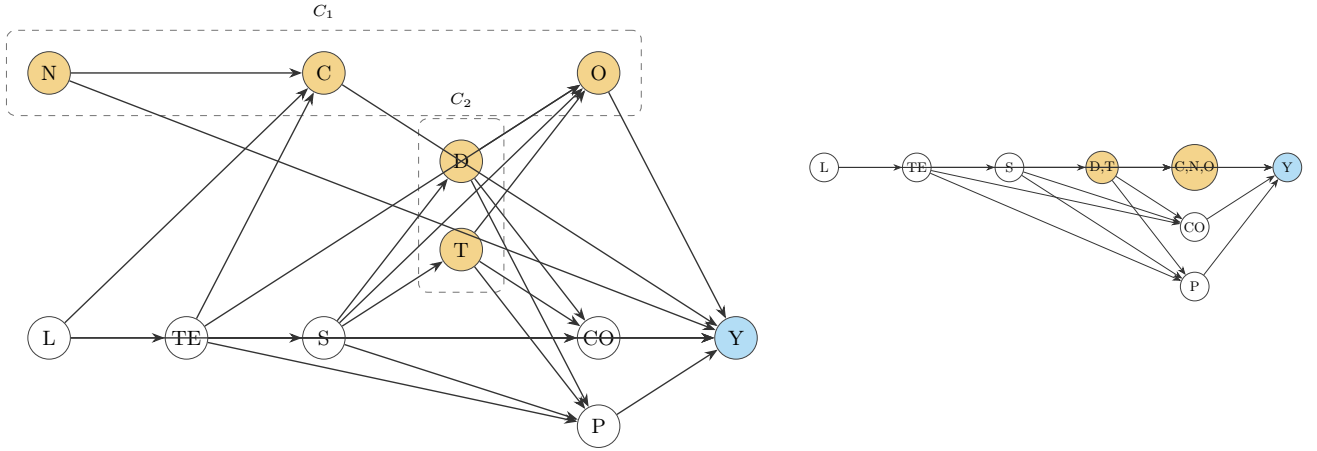


Figure 10: **SimplifiedCoralGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

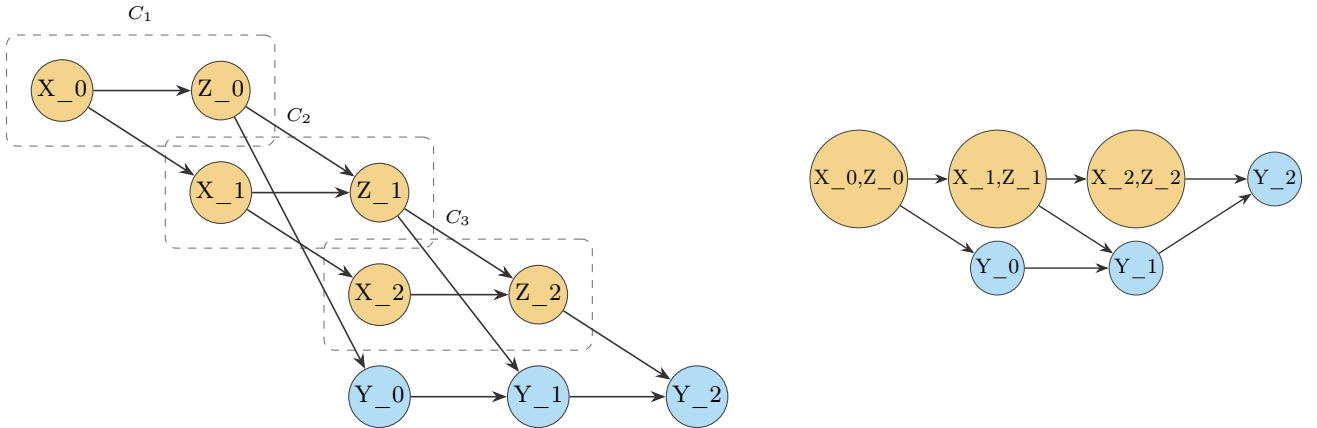


Figure 11: **DCBO stat / nonstat (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. Slices $t = 0, 1, 2$; the nonstat setup shares this topology with a change point in the SEM.

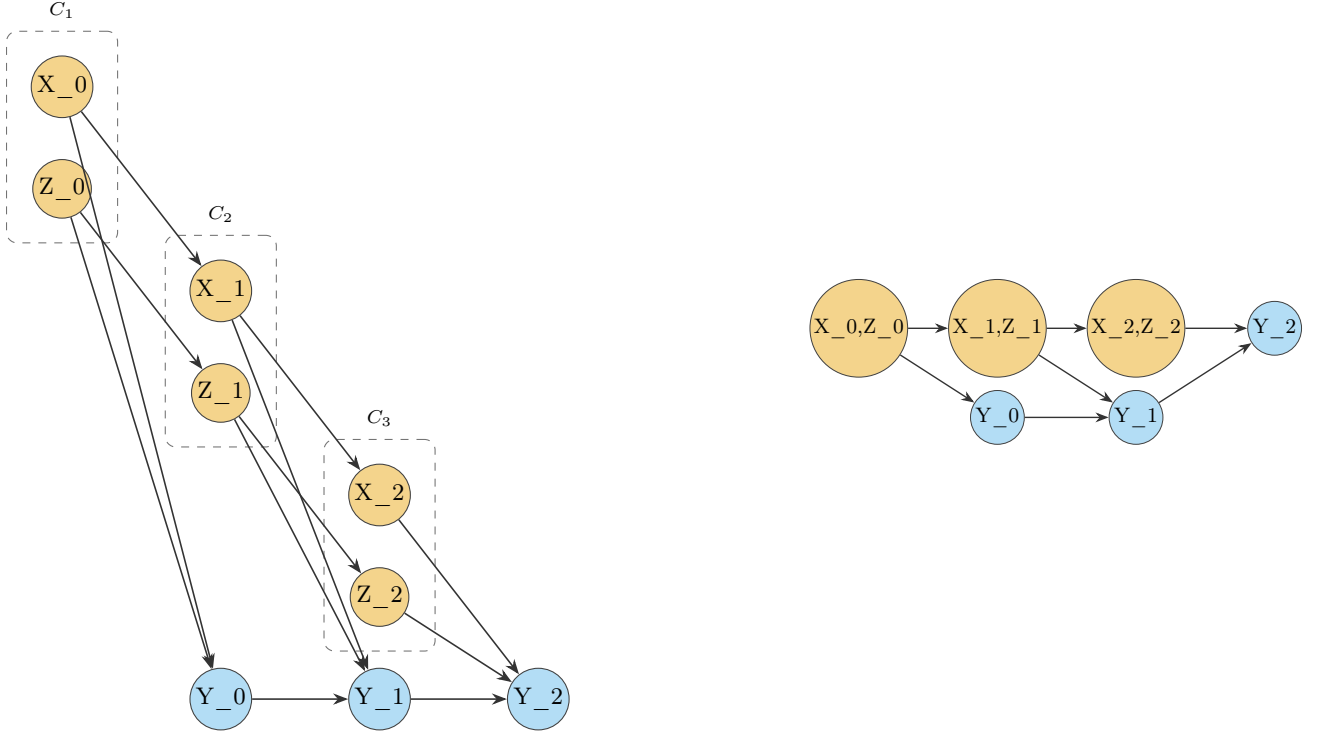


Figure 12: **DCBO ind (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.



Figure 13: **ToyGraph-M (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

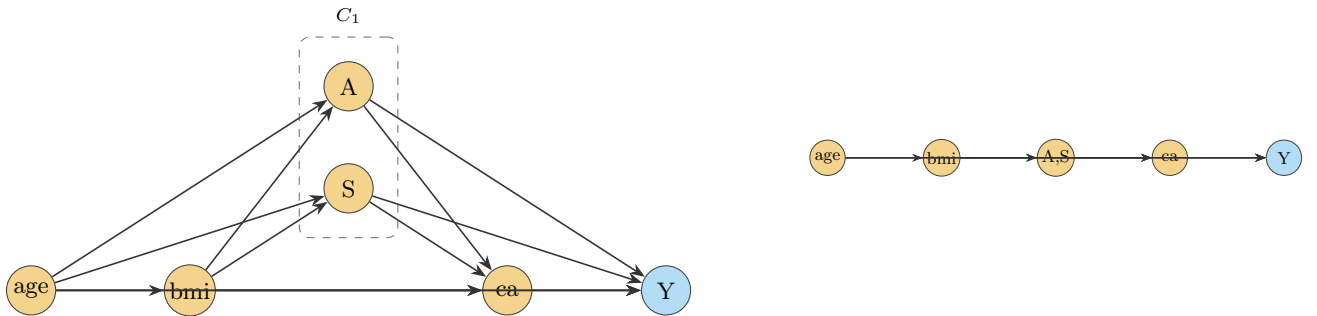


Figure 14: **PSAGraph (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

Algorithm 1 QCBO: Quotient Causal Bayesian Optimization

Require: C-DAG $\mathcal{G}^\Pi$ over the clusters of Π (or a fine hypothesis $\mathcal{G}$ and Π); target Y ; number of trials T ; observational data $\mathcal{D}^O$; optional hierarchy H (§3.1).

- 1: If given $\mathcal{G}$: $\mathcal{G}^* \leftarrow$ latent projection of $\mathcal{G}$; $\mathcal{G}^\Pi \leftarrow$ quotient of $\mathcal{G}^*$ under Π . Check acyclicity of $\mathcal{G}^\Pi$ (Assumption 1).
- 2: Build $\mathbf{ES}^\Pi \leftarrow \text{MIS}(\mathcal{G}^\Pi)$ via Eq. (4).
- 3: For each $\mathbf{X}_s \in \mathbf{ES}^\Pi$: run complete ID on $P(Y \mid \text{do}(\mathbf{X}_s))$ and fit the prior mean m_s via the two-tier rule, Eq. (5), on $\mathcal{D}^O$.
- 4: Collect initial interventional data $\mathcal{D}_0^I$.
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Update GP posteriors $(m_{s,t}, k_{s,t})$ for all $\mathbf{X}_s \in \mathbf{ES}^\Pi$.
- 7: $(\mathbf{X}_{s_t}, \mathbf{x}_t) \leftarrow \arg \max_{s, \mathbf{x}} \text{EI}_t^s(\mathbf{x}) / \text{Co}(\mathbf{X}_s, \mathbf{x})$.
- 8: Intervene, observe y_t ; update $\mathcal{D}_t^I$.
- 9: **if** H declared **and** incumbent plateaued: split the incumbent’s clusters via H , re-check validity, rebuild $\mathbf{ES}^\Pi \cup$ past arms, carry $\mathcal{D}_t^I$ by arm, init new arms (§3.1).
- 10: **end for**
- 11: **return** $(\mathbf{X}_{s_{t^*}}, \mathbf{x}_{t^*})$ where $t^* = \arg \min_t y_t$.

Table 3: QMCBO vs MCBO (its own harness and protocol; 5 seeds, $T=100$). Final Y (all tasks maximized) and wall-clock per unit.

Env	Method	Final Y	s/unit
ToyGraph ($y^*=2.172$)	MCBO	1.143 ± 0.271	1410
	QMCBO	2.150 ± 0.013	1570
PSAGraph ($y^* = -4.13$)	MCBO	-5.153 ± 0.000	6830
	QMCBO	-5.153 ± 0.000	2266

Table 4: Positioning of QCBO against the nearest methods.

Method	What is learned	Structural prior	Search space
CBO [Aglietti et al., 2020]	nothing (oracle DAG)	full DAG	MIS
GACBO [Mukherjee et al., 2024]	posterior over DAGs	DAG family	union of MIS
Durand et al. [Durand et al., 2025]	$\text{Pa}(Y) \subseteq \mathbf{X}$	Y leaf	$\mathbf{X}$
Park et al. [Park et al.]	$\text{PAG} \rightarrow \text{MAG}$	MEC	partial $\mathbf{X}$
SCB [Lee and Bareinboim, 2018]	nothing (oracle DAG)	full DAG	POMIS
QCBO (ours)	nothing	partition + C-DAG	identifiable cluster unions

Table 5: Invariance under an intra-cluster misspecification of the model’s graph view (same seeds, deterministic runs): trajectories identical to the correct-graph run, and mean $|\Delta_{\text{final}}|$. QMCBO cannot move (Theorem 1). MCBO does.

Env	Method	identical	$ \Delta_{\text{final}} $
ToyGraph	MCBO	2/5	0.299
	QMCBO	5/5	0.000
PSAGraph	MCBO	0/5	0.000
	QMCBO	5/5	0.000